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WORK ANALYSIS FOR PERSONNEL SELECTION

Antonio León García-Izquierdo

UNIVERSIDAD DE OVIEDO, SPAIN

Luis Díaz Vilela

UNIVERSIDAD DE LA LAGUNA, SPAIN

Silvia Moscoso

UNIVERSIDAD DE SANTIAGO DE COMPOSTELA, SPAIN

Introduction

Work analysis (WA) is by now a classic topic in I/O literature due to its long tradition and implementation in companies all over the world, particularly when dealing with personnel selection, promotion, or training. Consequently, WA could be seen as the cornerstone of human resources management because crucial decisions are based on information extracted from this type of report. However, the changing nature of jobs now means that human resources practitioners need a more active and updated approach to WA. The main reason for those changes is the huge development of information technologies, which constitute the origin of new jobs that are constantly emerging. These jobs are highly cognitively loaded and are appearing in all parts of the globe. At the same time, most traditional jobs have also been affected in some way by information technology, placing customers at the center and increasing the pressure for organizations to be online. In this way, duties and tasks are now more diverse and complex, meaning that WA faces new challenges if it is to maintain its relevance in the human resources arena.

In this chapter we use the concept of WA as proposed by Sánchez and Levine (2001) because it is a general concept that is useful for various purposes. As is commonly known, *job analysis* has been the most widely used term in the scientific literature, but WA can also encompass all the concepts that are concerned with the analysis of what people do at work. This facilitates the inclusion of different aspects of work such as tasks, abilities, and roles, and even the evolving nature of work over time and across organizations (Sánchez & Levine, 2012). The uses of WA are

manifold, but our objective in this chapter is to shed some light on the main issues around WA for personnel selection purposes.

Broadly speaking, we can define WA as the systematic process for gathering information about activities (e.g., tasks, duties, and responsibilities) and the personal attributes an employee must have in order to accomplish the objectives the organization has established for a particular job, through a procedure of analysis, synthesis, and inference. This analysis consists of subdividing jobs into more detailed elements to find out specific and important aspects for further applied decision-making purposes. Usually, the output of this analysis is a complete and detailed description of the job (i.e., job description) and the inferred workers' traits required for successful performance (i.e., job specification). Consequently, we can use WA for both criteria and predictor development, and for calculating the degree of alignment between criteria and predictors. Another key and related issue is the question of person-work fit, a topic that we are going to deal with in detail in this chapter.

Originally, WA was focused on a standard protocol, and it can be traced back to the principles of scientific management in the early years of the twentieth century, with the study of times and movements by Taylor, and by Lillian and Frank Gilbreth, who were the pioneers in the analysis of workflow from a rational and empirical point of view. Some of the forefathers of personnel selection from different parts of Europe, such as Lahy (France), Lipmann and Stern (Germany), and Mira (Spain), made relevant contributions to WA as well (Salgado, Anderson, & Hülsheger, 2010).

Work analysis implies some costs in terms of time and money, so the effort made to develop a well-designed WA project depends on the gains expected from this process. In this sense, jobs that require high cognitive demands, those full-time jobs with open-ended contracts that are long-term oriented and frequently in the upper part of the organization, more easily justify a WA process despite being the most difficult to analyse. On the other hand, part-time and temporary jobs, which are usually less cognitively demanding, are less likely to deserve an exhaustive examination. The information derived from WA can be classified into three categories: job identification, job description, and job specification. Job identification includes information about job titles and physical and/or functional location. Job description includes a job summary and the workflow breakdown into duties and tasks performed by the incumbents. Other relevant information such as job conditions, methods, or techniques can also be included in the job description. Job specification refers to job requirements needed to carry out the job's duties and tasks. It may include the knowledge, skills, abilities, and other characteristics (KSAOs) that are used in personnel selection procedures, such as memory or initiative (Brannick, Levine, & Morgeson, 2007). Nonetheless, from both a theoretical and a practical perspective, some authors and practitioners defend different and apparently opposed points of view. In the next section, we are going to deal with the classical dilemma about the orientation of the WA, that is, towards the tasks or towards the workers.

Task-oriented analysis and worker-oriented analysis

Following Gael (1988) and Brannick and Levine (2002), we can distinguish between task-oriented methods and worker-oriented methods of WA.

Task-oriented job analysis

This approach considers duties, functions, and tasks as the basis for studying jobs in the workplace. Consequently, particular employees' attributes are not of interest, as worker characteristics display variability in how they perform those tasks depending on their experience, knowledge, abilities, personality, and other attributes. The task-oriented approach places the emphasis on the structure of the organization, which remains relatively stable. From this perspective, jobs are independent of the people who perform them. The best-known methods are the following: Time and Motions Study, Functional Job Analysis (Fine, 1988), Task Inventory/ Comprehensive Occupational Data Program (TI/CODAP), Work Performance Survey System, and the Critical Incident Technique (Flanagan, 1954). Particularly important is the Functional Job Analysis, which is used in the development of the Occupational Information Network (O*NET) database, known for its well-structured synthesis perspective for most jobs. When dealing with very important decisions in terms of money or consequences for people or resources, the Critical Incident Technique is an appropriate method.

Worker-oriented job analysis

The worker-oriented analysis approach stresses the individual characteristics an employee must have for adequate performance. We can differentiate between two types of these characteristics: traits and competencies. Traits are based on stable characteristics, such as aptitudes and personality. Although the concept of competence is far from being clear, we can see competencies as those trainable core characteristics, in terms of behaviors, that lead employees to success (Boyatzis, 1982; McClelland, 1973). These are mainly KSAOs, such as attitudes and motivation (Campbell, McCloy, Oppler, & Sager, 1993). Due to the existence of a growing consensus around the competency model, research on worker traits has suffered a dramatic reduction. The fast and continual changes occurring in the content of jobs and the consequent instability in terms of tasks are impelling a growing number of modern organizations to start using Competence Modelling (CM) in their human resources practices (Schippmann, 1999). The main characteristic of CM is that it enables the identification of KSAOs that tend to be broad and that are not necessarily linked to specific jobs or tasks (Brannick, Levine, & Morgeson, 2007). Therefore, CM takes into account organizational goals and missions and seeks to develop a set of competencies to be applied across the organization or units within the organization (Lawler, 1994). Regarding this, Goldstein and Ford (2002, p. 272) stated that “. . . these more global competencies are expected to not only predict behavior across a wide variety of tasks and settings but also provide the organization with

a set of core characteristics that distinguish the company from others in terms of how it operates its business and treats its employees.” Thus, CM approaches seem to offer the advantage of going beyond the information provided by traditional worker-oriented job analysis, which focuses on the identification of the most relevant KSAOs for the job. One of the main criticisms of CM has to do with the multiple and sometimes contradictory definitions of competence (Alliger et al., 2012; Schippmann et al., 2000; Voskuil, 2005). Nevertheless, some authors agree on considering the competencies as part of job performance (Bartram, 2005; Lievens et al., 2010). As Sánchez and Levine (2009, p. 61) put it,

In essence, whereas traditional job analysis focuses on describing and measuring the requirements of work, CM creates a conduit to influence day-to-day employee performance along strategic lines. Having highlighted how we stand to gain from using them together, we hope to stimulate research on the manner in which Task Job Analysis (TJA) and CM may supplement each other in a host of HR applications.

Sources of information

The most commonly used sources of information in the analyses have been job incumbents, although supervisors, peers, and job analysts are also frequently used as information agents. Recent changes in working conditions and arrangements are blurring job boundaries and emphasizing teamwork, making it necessary to broaden the number and type of informants, including internal and external customers, and the number of activities observed (Pearlman & Sánchez, 2010; Voskuil, 2005). This changing nature of work makes WA even more important now (Guder, 2012) and increases the necessity of performing analyses systematically.

As Guder (2012) points out, the blurring of job boundaries entails an increasing need for coordination of activities between different jobs, that is, a “crossfunctional coordination of activities” (p. 37). This implies the appearance of a new form of cross-supervision, not necessarily hierarchical in nature, as it could also take the form of an internal customer. This is probably not new, but it has been receiving particular attention recently (e.g., Sánchez & Levine, 2012). This approach would make WA even more complex because it would require the participation of incumbents, direct supervisors, and coworkers whose jobs are related to the target job.

One more effective way of putting together all the possible WA data could be through a panel of experts who would catalyse all the work information from the incumbent, the analyst, and any other source. A panel of this type should include experts on WA as a technique and experts on the organization and on the target job under analysis. This panel should have different meetings before and after each step, making decisions about what is valid and what is not. As Wernimont (1988) states, using panels of experts will add reliability to a process currently based around one individual, without underestimating the importance of the participation of analysts too.

Methods for WA

This section is devoted to the methods for gathering information through the WA process. Pearlman and Sánchez (2010) describe four categories of WA applications: predictor development, criterion development, validity evidence development, and validity evidence extension. The first category constitutes the most used application, and its goal is to describe jobs in terms of the KSAOs needed to perform them properly. That is, when WA is used for selection purposes, a job description is not enough, and job specifications are necessary too. These specifications require KSAOs, and they are to be used as job performance predictors. Therefore, they constitute decision-making criteria affecting candidates' lives as well as organizational effectiveness, and they should have adequate levels of reliability, validity, discriminability, and practicality (Ployhart, Schneider, & Schmitt, 2006). These characteristics should be expected not only for the instruments used but also for the entire process of WA and the derivation of KSAO profiles.

WA procedures should be iterative: departing from observation, a description should be generated that, in turn, should be confronted with reality, reformulated when necessary, and confronted again. This procedure could last forever, but the intervention of several agents in different points considerably reduces the number of iterations. Each phase in WA deserves special attention and care. We need to point out that even when our goal is to obtain a job-specific KSAO profile, the preliminary steps of observation and description of work activities and behaviors are both necessary.

There are two classic approaches to classifying the different methods and instruments for WA: the qualitative and the quantitative. The qualitative approach is based on gathering information through techniques such as interviews or focus groups, where job incumbents provide ample narrative information stressing not only the content of the job but the context as well. On the other hand, the quantitative approach is mainly based on structured questionnaires where job incumbents are strictly restricted to limited answering options, or even using the organization's work results records (number of clients, money, production, accidents, etc.).

Qualitative methods

Observation

WA is an observational process. A work analyst resembles an ornithologist trying to describe an eagle's behavioral pattern through the seasons, with the only difference being that incumbents can think and speak. But these characteristics of the observed performer do not free the analyst from the need for thorough observation of the performance, whatever method and instruments are used. In fact, observation is one of these methods and is frequently ignored. Observation has four advantages over other methods (Martinko, 1988): (1) the first-hand nature of the data, as the information is obtained directly, (2) the opportunity to obtain in-depth qualitative information, (3) the improvement of face validity by providing organizational

jargon to the analysts and to the WA reports, which also improves confidence, (4) information can be purely descriptive, as well as serving inferential hypothesis testing purposes. At the same time, observation also has some disadvantages, which Martinko (1988) describes as time, cost, extension of reports, coordination requirements, incumbents' reactivity, and difficulty in capturing non-observable behaviors. Despite these shortcomings, observation is advisable in the first stages of WA because it will provide the analyst with helpful knowledge about the job and will give the required information for asking appropriate questions in the subsequent interview (McPhail, Jeanneret, McCormick, & Mecham, 1989). Finally, observation becomes necessary particularly when there is a lack of consensus and the panel of experts cannot reach a solution.

Interview

The interview can be defined as a meeting between two or more people for the purpose of exchanging information about a job or a series of jobs (Gael, 1988). Analysts should be trained interviewers, and interviews should be well planned in order to obtain enough quality information to accomplish WA goals. When planning an interview, Gael (1988) suggests four considerations: setting explicit objectives, identifying the class of interviewees, specifying the degree of structure that the interview will have and the number of interviewees, and preparing the necessary materials and equipment. Interviews can and should be performed in different stages in WA in order to obtain (*initial interview*) and refine (*verification interview*) information about the work performed. There are a number of publications on how to conduct WA interviews, from Gael's (1988) chapter to, for instance, the practical guide to interviews by the U.S. Office of Personnel Management (2008). Besides this, McPhail et al. (1989) also provide some suggestions for improving interview effectiveness, including the selection of location, the preparation and structuring of the interview, and the scheduling arrangement with the interviewee. Although the selection interview should be structured, in the light of previous research (Motowidlo et al., 1992), the initial WA interview should be unstructured because its purpose is to obtain as much work-related information as possible, while verification interviews should be structured and centered on the inconsistencies or discrepancies found.

Quantitative methods

Job-task inventories

Job-task inventories are basically questionnaires composed of a background information section followed by a task inventory section, that is, a list of tasks arranged by general functions or duties (Christal & Weissmuller, 1988). Tasks are components of jobs and can be described as a set of activities oriented to accomplishing a single goal (Gillan, 2012; Voskuil, 2005). They are formulated using an action verb, an object of the action, and essential modifiers. For example, a clerical task could be

“Prepare documents for verification and signature of the superior”. Task lists are generated by incumbents, supervisors, analysts, a Subject Matter Experts (SME) board, or any combination of these. Once the list is complete, a questionnaire is made using tasks as stimuli, and a Likert-type scale with a set of anchorages for importance, frequency, criticality, and/or complexity. In fact, there will be as many task inventories as types of scales used (one for importance of tasks, one for frequency, and so forth). These questionnaires can be administered to incumbents or to anyone closely familiar with the work under analysis. As a result, the analyst will have an image of the work done in the form of a list of tasks with their associated properties (frequency, etc.), so that core tasks in the job can be easily identified. This is a work description technique and is intended as an initial step in the development of work specifications.

Fleishman Job Analysis Survey (F-JAS)

The Fleishman Job Analysis Survey (F-JAS) is a system for analysing the KSAOs needed to perform jobs (Management Research Institute, 2013). The F-JAS is a 73-item questionnaire measuring three domains of required KSAOs: cognitive, psychomotor, and physical. Within the first domain, seven cognitive ability requirements are measured: verbal, idea generation and reasoning, quantitative, memory, perceptual, spatial, and attentiveness abilities. With regard to psychomotor abilities, F-JAS measures fine manipulative, control movement, and reaction time and speed abilities. Finally, requirements of five physical abilities are measured: physical strength, endurance, flexibility, visual, and auditory and speech abilities. Contrary to its apparent simplicity, the F-JAS is not, in fact, particularly straightforward to implement. Caughron, Mumford, and Fleishman (2012) recommend four steps to be followed in its application: (1) reviewing job description materials and the F-JAS Administrator's Guide. When these materials do not exist, observation, interviews, and/or job inventories should be undertaken; (2) creating a panel, with the suggestion of about 20 SMEs to obtain sufficient levels of interrater reliability; (3) having each SME rate jobs using the questionnaire. Raters must decide whether each ability is required, and if so, to what extent the ability is required; (4) determining mean ability scores obtained from the SMEs for each job, requiring consensus meetings when interrater reliabilities are low.

Practitioners should be aware of the subjective nature of F-JAS data and the reliability threats discussed above. In addition, special attention should be paid to the difficulties that applicants in particular have in separating their own abilities and levels from those required to perform their jobs.

Personality-based job analysis

Several authors have developed instruments with the aim of evaluating personality-related job requirements. Hogan and Rybicki (1998) developed the Performance Improvement Characteristics Job Analysis (PIC) for use in selection and

TABLE 1.1 Questionnaire examples of Performance Improvement Characteristics Job Analysis (adapted from Hogan & Rybicki, 1998)

<i>Would job performance improve if the incumbent _____?</i>	
- Is steady under pressure	_____
- Is not easily irritated by others	_____
- Is competitive	_____
- Takes initiative – solves problems on his/her own	_____
- Needs variety at work	_____
- Is kind and considerate	_____
- Supports the organization’s values	_____
- Is imaginative and open-minded	_____
- Remembers details	_____
<i>(Response Options: 0. Does NOT Improve Performance; 1. Minimally Improves Performance; 2. Moderately Improves Performance; 3. Substantially Improves Performance)</i>	

development. The PIC is a structured questionnaire based on the Five-Factor Model (FFM) of personality (Digman, 1990) and, specifically, on the scales of the Hogan Personality Inventory (HPI; Hogan & Hogan, 2007). The PIC consists of 48 items, which are evaluated using four-point scales that compound seven dimensions in which the most important personality characteristics for the evaluated job are included (Adjustment, Ambition, Sociability, Interpersonal Sensitivity, Prudence, Inquisitive, and Learning Approach). The PIC shows acceptable psychometric properties (Hogan & Rybicki, 1998). In Table 1.1 some of the PIC items are shown.

Another personality-oriented WA instrument is the Personality-Related Position Requirements Form (PPRF; Raymark, Schmit, & Guion, 1997). This form includes 107 items and evaluates 12 personality dimensions, which are based on the FFM: (1) General Leadership, (2) Interest in Negotiation, (3) Ambition (these three dimensions are included within the Extraversion Factor of the FFM), (4) Friendly Disposition, (5) Sensitivity to Interest of Others, (6) Cooperative or Collaborative Work Tendency (these two dimensions are included within the Agreeableness Factor), (7) General Trustworthiness, (8) Adherence to a Work Ethic, (9) Thoroughness and Attentiveness to Details (Conscientiousness Factor), (10) Emotional Stability, (11) Desire to Generate Ideas, and (12) Tendency to Think Things Through (Intellect Factor).

Raymark et al. (1997), using descriptions of 260 different jobs and 12 occupational categories, found evidence that the PPRF reliably differentiates between jobs. They found interrater coefficients ranging from .66 to .92 for the different occupational categories. Their results also showed acceptable reliability for most dimensions. The alpha coefficients ranged from .72 to .92. Table 1.2 includes some items of the PPRF.

TABLE 1.2 Examples of items on the Personality-Related Position Requirements Form (adapted from Raymark et al., 1997)

Effective performance in this position requires the person to:

- Take control in group situations
 - Mediate conflict situations without taking sides
 - Seek challenging tasks
 - Start conversations with strangers easily
 - Give constructive criticisms tactfully
 - Work as part of an interacting work group
 - Have access to confidential information
 - Meet specified deadlines
 - Keep cool when confronted with conflicts
 - Find new ways to improve the way work is done
 - Identify and evaluate options before taking action
-

(Response Options: Not Required; Helpful; Essential)

Position Analysis Questionnaire (PAQ)

This instrument can be defined as “a structured job analysis questionnaire which provides for a quantified analysis of individual jobs in terms of a number of ‘job elements’, often referred to as ‘items’” (McPhail et al., 1989, p. 1). The PAQ consists of 195 items. Of these, 187 relate to job activities or to the work environment; the remaining eight items are used to report the type of compensation (PAQ Services, 2013). It is a worker-oriented questionnaire that is intended to describe jobs based on six domains: information sources, mental processes, responses or actions involved in jobs, interpersonal activities in work, work situation and context, and other miscellaneous aspects of work (McCormick, Mecham, & Jeanneret, 1977). Job elements are arranged within these six divisions. Each division contains several factors. These authors described the process of factor analysis that gave 32 division dimensions and 13 overall dimensions. In short, the first 32 dimensions resulted from six division-specific factor analyses, while the 13 overall dimensions are the result of a factor analysis on all the items together. Following PAQ Services (2013), job component validity analyses with the General Aptitude Test Battery (GATB) makes the PAQ useful in identifying appropriate selection tests and cut-off scores, and providing job component validity evidence for ability tests.

Although the Position Analysis Questionnaire has received attention in almost every work/job analysis handbook, little has been said about its application procedure. Following Mecham, McCormick, and Jeanneret (1977), the use of the *Job Analysis Manual* is strongly recommended when completing the questionnaire. As an example, in an unpublished part of a study, data from two experienced analysts showed that using (or not using) the manual caused clear differences in results (Díaz Vilela et al., 2008). These differences were eliminated when both analysts used the manual to answer the questionnaire.

WA databases

Currently, since the Dictionary of Occupational Titles (DOT) has become out-dated, the most comprehensive database of this type is the O*NET (Peterson et al., 1999). It is a dynamic system of information, services, tools, and applications for more than 800 occupations that are updated regularly. The system provides occupational profiles filled with information regarding the required knowledge, skills, abilities, work activities, work styles, work interests, background, education and training requirements, and work context. The O*NET content model is organized on the basis of six sets of descriptors: (1) worker characteristics, (2) worker requirements, (3) experience requirements, (4) occupational requirements, (5) occupation-specific requirements, and (6) workforce characteristics. The first three sets are worker-oriented, and the last three sets are job-oriented. Brannick, Levine, and Morgeson (2007) made a complete and comprehensive summary of these descriptors and their categories:

1. Worker characteristics are personal attributes needed for successful job performance. They are divided into three categories: (a) abilities (e.g., oral expression, originality, or memorization), (b) occupational interests and values (e.g., achievement, creativity, or moral values), and (c) work styles (e.g., persistence, cooperation, or integrity).
2. Worker requirements are personal attributes that are developed through experience and that are helpful in performing different tasks. They are composed of three categories: (a) basic and cross-functional skills (e.g., mathematics, reading comprehension, or persuasion), (b) knowledge (e.g., psychology, music, or art), and (c) education (e.g., general mental ability or instructional program).
3. Experience requirements include specific vocational training, work experience, and licensing (e.g., subject area education, licenses required, requirement to obtain license, or related work experience).
4. Occupational requirements are referred to as what the worker does. This descriptor is composed of three categories: (a) generalized work activities (e.g., getting information needed to do the job, making decisions and solving problems, or documenting and recording information), (b) work context (social interaction, radiation exposure, or level of automation), and (c) organizational context (e.g., empowerment, task identity, or skill variety).
5. Occupation-specific requirements refer to some elements that apply to a single occupation or a narrow job family. It includes specific tasks, tools, and technology.
6. Workforce characteristics are not incorporated directly into O*NET. This category is designed to link to other databases with relevant information. This descriptor includes labor market information, occupational outlook, and wages.

Besides offering a conceptual model, O*NET also provides other useful tools and resources, such as O*NET OnLine and O*NET Questionnaires. O*NET OnLine is a web-based application that allows users to search and access occupational data. O*NET Questionnaires are specific occupational analysis instruments

which refer to knowledge, skills, abilities, work activities, work context, background, education and training, and work styles.

Another classification is the ISCO-08 (International Standard Classification of Occupations), which replaced the ISCO-88 and was endorsed by the Governing Body of the International Labour Organization (ILO) in March 2008. This is a tool for organizing jobs into a clearly defined set of groups according to the tasks involved in each job. The occupational classification system includes major, sub-major, minor, and unit groups of the different occupations and the descriptions of each one.

Reliability and validity in WA

Reliability and validity are complex domains that depend on a number of factors. At the very least, raters' own KSAOs (Morgeson & Campion, 1997; Sánchez & Levine, 1994), relative position of the informant/information complexity (Dierdorff & Wilson, 2003; Sánchez, Zamora, & Viswesvaran, 1997), and even demographic background (Arvey, Passino, & Lounsbury, 1977) should be taken into account. These factors affect the quality of the data because they produce informational distortion, which is difficult to reduce. Morgeson and Campion (1997) proposed a list of 16 sources of inaccuracies grouped in two categories, social and cognitive. Social sources are social influence, group polarization, motivation loss, impression management, social desirability, and demand effects. Cognitive sources are information overload, heuristics, categorization, carelessness, extraneous information, inadequate information, order and contrast effects, halo, leniency and severity, and method effects (Morgeson & Campion, 1997).

The conceptualization of validity and, particularly, reliability, is an important topic that should be stressed at this point. Reliability refers to the consistency between raters or within raters (Sánchez & Levine, 2012). Such a clear, simple definition conceals a quite complex concept and certain methodological problems. First, consistency between raters has been represented through intraclass correlations, assuming that high correlations between raters' scores indicate agreement. On the other hand, high correlations between test and retest application to the same rater is assumed to show stability in this rater's observations (Dierdorff & Wilson, 2003; Viswesvaran, Ones, & Schmidt, 1996). Yet neither of these assumptions is rigorously true. Starting with the latter, stability within a rater's scores in a test-retest situation may be due to individual biases when a job has changed in nature: different observed reality – same informed scores. In other cases, observed instability can be a raw reflection of reality and therefore have high reliability. In these cases, stability or consistency is not a reliability indicator.

With respect to interrater consistency, correlations are good indicators of the existence of a relationship and of its strength, but only that. Consistency is not agreement (Ployhart, Schneider, & Schmitt, 2006). Interrater correlations with values around $r_{xy} = .90$ may come from appraisers whose means are quite different. When a practitioner trusts interrater stability, as often happens, the assumption is that the mean scores in each criterion should guide the establishment of the cut-off

score for selection purposes. Regardless of which analyst is right, it remains clear that the mean represents none of them.

These are real threats to validity and hence to the quality of WA and subsequent criteria development (Harvey, 2012). For this reason, the concept of reliability in WA should be extended to capture not only consistency, but agreement too. This assertion is especially oriented to practitioners in real-life settings, where often the time and economic expense of analysing a statistically adequate number of positions, or the hiring of a number of analysts, is unaffordable. When this can be done, a cluster analysis of variables should result in as many clusters as job facets. A cluster analysis of subjects (analysts) should result in only one cluster when all the analysts agree. In reliability terminology (Revelle, 2013; Revelle & Zinbarg, 2009), the total omega should be high and the hierarchical omega should be low when variables are analysed, indicating the existence of several underlying factors (job facets); but the hierarchical omega should be high and the total omega low when subjects are treated as variables, indicating the existence of a unique underlying component (agreement among analysts). So, what is the solution to this dilemma? Some decades ago, Wernimont (1988) proposed six guarantees of content validity in job analysis oriented to selection procedures: (1) familiarity with the job analysed, which is expected from incumbents and supervisors and should be gained by analysts through observation and/or interpretation of data; (2) the use of a panel of experts instead of a single informant; (3) training observers in behavioral sciences, although this alone is not sufficient; (4) expertise in tests and attribute derivation from WA; (5) knowledge of existing tasks and attribute taxonomies; and (6) after a WA procedure, relationships among tasks and among attributes should show content validity by themselves. Following these guidelines, a WA should start with job familiarization on the analysts' part, and technical familiarization on the experts' part. There is not much literature on how to train experts in the use and interpretation of analysis tools, so training programs should be developed *ad hoc*.

The question of fit

Job fit is a concept that explains whether the intersection between an employee's characteristics and the requirements of a particular job and work environment match or not. The use of concepts like work roles, team-person fit, and person-organization fit (Lievens, Highhouse, & De Corte, 2005) has been suggested as a substitute of the analysis centered on jobs. Anyway, we can distinguish between demands-abilities (competencies) fit and needs/values-supplies fit. When the two interests match, an employee and an organization experience a good job fit. A good job fit coincides with high levels of job and organizational performance and job satisfaction. The classical approach in personnel selection has put the emphasis on matching individuals and jobs (for instance, Lofquist & Dawis, 1969), and even more recently Kristof-Brown, Zimmerman, and Johnson (2005, p. 281) conceptualized fit as "the compatibility between an individual and a work environment that occurs when their characteristics are well matched". However, nowadays the interest has shifted to group and organizational fit. In this sense, it is important to capture

some organizational factors within which the work is carried out, such as leadership, strategic planning, focus on customer satisfaction, and emphasis on quality and productivity (Offermann & Gowing, 1993). This could include sharing team or organizational values, expectations, and climate. The organizational level is mainly related to organizational culture, which is made up of internal values, norm beliefs, underlying assumptions, attitudes, and behaviors shared by a group of people. Culture can be observed throughout the behavior that results when a group arrives at a set of – generally unspoken and unwritten – rules for working together. We can see the organizational culture by means of language, symbols, legends, decision making, and work routines. Person-organization fit deals with the compatibility of the individuals and the organization. Muchinsky and Monahan (1987) distinguished between complementary and supplementary fit. Complementary fit occurs when individuals fill a gap in the environment (i.e., the job) or vice versa, whilst supplementary fit is when the individual and the environment are similar. When looking for applicants to match a specific organizational culture, the steps are similar to the process at the individual level. The main difference between the organizational and the individual level is that the former put the emphasis on the lifestyle-world views. Consequently, when analysing candidates to match the organizational culture, the main issues involve personal values, attitudes, and personality.

WA and equal employment opportunities

Finally, we would like to stress the social dimension of fit and WA. When conducting a personnel selection process, it is necessary to take into account the social and legal requirements as well as the individuals and the organizations. This has been brought to the foreground through the concept of Equal Employment Opportunities (EEO). EEO deals with fairness in an increasingly diverse workforce, in terms of demographic issues such as age distribution and the sexual, ethnic, and cultural composition of the workforce. The challenge is about job-relatedness, so organizations must document some linkage between KSAOs and job tasks. However, there must be thoroughness in the coverage of such KSAOs, focusing on the whole person, avoiding invasiveness, and using unbiased instruments.

As Biddle and Kuthy (2012, p. 365) state, a rigorous WA “provides crucial data for ensuring that EEO objectives are being met”. This means that if the WA and work specification are inaccurate, some qualified people, sometimes protected groups, may be excluded from succeeding. Consequently, the selection process could be unfair. The European Union (EU) has protected equality in the workplace, developing extensive legislation to prevent discrimination, especially between women and men. This equal-opportunity legislation arose in the seventies with the enactment of some relevant directives such as Council Directive 75/117/EC regarding equal pay and Council Directive 76/207/EEC on equal treatment in employment and vocational training. At the beginning of the present century, there was a renewed attempt to deal with this matter in the shape of Council Directive 2000/78/EC, which established a framework to prevent all types of discrimination. Two years later, Parliament and Council Directive 2002/73/EC amended

Council Directive 76/207/EEC, giving Member States an active role in achieving the objective of equality between men and women when formulating and implementing laws, regulations, administrative provisions, policies, and activities in employment and vocational training.

As stated in EU Directive 2006/54/EC, discrimination can be performed directly, if one person is treated less favorably on grounds of gender, or indirectly, if an apparently neutral provision, criterion, or practice would put persons of one gender at a particular disadvantage compared with those of the opposite gender. Indirect discrimination alone can be acceptable only in the case of the following three assumptions: (1) for professional activities where gender is a factor because of the nature or conditions of the activity to be performed, (2) to protect women, particularly in pregnancy and childbirth, and (3) to promote equal opportunities between men and women (art. 2.6 EU Directive 76/207 on Equal Treatment). Considering all this legislation, we can conclude that decision making in personnel selection should be based on those criteria clearly related with an accurate and thorough WA. That is, the personnel selection decision being made must be based on tools and instruments that present enough evidence of criterion-oriented validity. That means that WA is not only a useful tool in human resource management but is also essential in avoiding discrimination in the workplace.

Recruitment and WA

The inherent relevance of WA, and more specifically of job descriptions, for recruitment purposes demands a specific section in the present chapter. In fact, the job description very often constitutes the basis of the recruitment advertisement and establishes the ways by which the information will appear reflected. As such, it affects the degree of attraction of the recruitment advertisement that is perceived by the potential candidates and, consequently, has important implications for the success of the entire selection process and for the fulfillment of the job vacancy.

Hurtz and Wright (2012) describe how WA can be used to maximize the effectiveness of recruitment, and, therefore, to improve the utility of selection processes, optimizing the percentage of high-quality applicants (highly suitable applicants) that enter the selection process and remain interested in it, and, consequently, stay in the organization. To achieve this goal the authors recommend the use of established taxonomies, because they provide a common and accepted language to describe jobs. On the other hand, the use of informational systems and communication technologies displayed by these taxonomies allows information to be obtained with more substance and fidelity about the job, compared with more traditional methods.

In this sense, Cober et al. (2003) indicate that the organization must focus on the content as well as on the style (e.g., usability, interactivity) of the information used in the recruitment. Hurtz and Wright (2012) provide a framework with four “layers” of work description that can be used as a guide for the use of WA in the recruitment process. The first layer refers to traditional work descriptions and details how the other layers can be built departing from it. The second layer adds an electronic database to the first to organize and disseminate job

descriptions, enabling users to download the respective information. The third layer adds color elements, graphics, and interactivity options to the work descriptions (e.g., hyperlinks), which provide additional information on job descriptions. The fourth layer allows the inclusion of multimedia components (e.g., audio, video, and two-way communication technologies such as message boards or online chat tools). Moreover, these authors stress that the use of this framework can make a valuable contribution in the accomplishment of the three main recruitment objectives: (1) attracting appropriate applicants; (2) maintaining their interest in the job offer and in the organization; (3) persuading applicants to accept the job offer. In any case, the basis of this model and of their four layers lies in the information about the job that is provided by the WA.

Conclusions

WA has a long history that has brought researchers and practitioners a handful of well-designed and useful techniques and instruments, which have been adapted to different times and contexts. Consequently, WA has had to cope with different challenges. One of the most important debates is around WA orientation, that is, task-oriented versus worker-oriented WA. From our point of view, both orientations are necessary, and the choice depends on the WA purposes. For instance, when grappling with personnel selection, worker orientation is unavoidable, but many times, they are complementary perspectives.

Simultaneously, models and principles of WA are changing, as everything is changing in the work arena nowadays. Particularly important are the effects of information technologies, as they are placing individuals working worldwide through connectivity. Consequently, jobs, roles, and performance are progressively far from a static place in the organizations and are more related with a dynamic relationship (García-Izquierdo, Ramos-Villagrasa, & Navarro, 2012). Moreover, workers are coping with the necessity of continuous job crafting (Tims, Bakker, & Derks, 2014) to develop their roles and to enhance their performance, in order to maintain their positions in an organization and across organizations eventually. The workforce is making a huge effort of redesigning jobs and professional competencies in a constant pressure to update and respond to organizational and job market demands. This has turned the view from job fit to organizational fit, and in many cases to job market fit, in order to be employable. Organizations are shaping jobs' functions and their structure to the market demands, and simultaneously need to comply with legal restrictions, particularly in the context of diversity and equal employment opportunities. Thus, nowadays WA remains the core of HR policies, and a good (if not necessary) basis for relevant organizational issues. Even more, WA contributes to enhance responsibility assumed by incumbents and to reduce ambiguity when organizing tasks, and it plays an important role in goals achievement indeed.

All in all, the present chapter suggests that WA is necessary for successful recruitment, personnel selection, and promotion processes. WA may also be one of the best means for guaranteeing equality and avoiding discrimination in the workplace.

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